



Engr. Abdul Wahab

Mechanical Design Engineer & Product Designer

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Upwork	upwork.com/freelancers/abdulwahab	Portfolio	wahababdull.github.io/CV/

PROFILE SUMMARY

Research-focused Mechanical Design Engineer with a strong background in computational mechanics (FEA/CFD) and kinematic mechanism design. I have worked on advanced simulations and parametric modeling involving autonomous mechanical systems, unmanned aerial vehicles (UAVs), and fluid-mechanical systems. With over 2 years of professional experience in design, analysis, mechanical automation, and additive manufacturing, I am excited to advance my education and contribute to cutting-edge research in Unmanned Aircraft Systems (UAS) and morphing wing kinematics.

RESEARCH INTERESTS

Consumer Products	Unmanned Aerial Vehicle (UAV) Airframe Design
Unmanned Aircraft System (UAS)	Autonomous Mechanical Systems Design
Morphing Wing Kinematics	Computational Mechanics, FEA, CFD
Additive Manufacturing, 3D Printing	

WORK EXPERIENCE

June 2026 — Present

Mechanical Design Engineer

Dysol (Pvt) Ltd, Lahore, Pakistan

R&D Department — Key Responsibilities & Contributions

- Serving as a core Mechanical Design Engineer in the Research & Development department, focusing on innovative product development.
- Leading the end-to-end design and optimization of complex mechanical systems, including medical devices and fitness equipment.
- Conducting iterative engineering analysis, simulation, and validation to ensure functional performance and reliability.
- Applying DFM and DFA principles to bridge the gap between conceptual design and full-scale manufacturing.
- Collaborating across disciplines to integrate mechanical, fluidic, and electronic subsystems into cohesive, production-ready architectures.

August 2023 — Present

Mechanical Design Engineer

Upwork (National & International Clients)

- Delivered professional mechanical and product design services to national and international clients, applying rigorous engineering standards.
- Engineered precise 3D CAD models, complex assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.

- Optimized components for consumer products and industrial applications, focusing on optimal material selection and manufacturing processes.
- Maintained a high degree of technical communication to translate client requirements into verifiable engineering parameters.

EDUCATION & TRAINING

September 2019 — October 2023

B.Sc. Mechanical Engineering

University of Engineering and Technology (UET), Peshawar — KPK, Pakistan

FINAL YEAR THESIS PROJECT

Design and Fabrication of Mechanism for Automated Installation of 18-Liter Water Bottle for Water Dispensers

Supervised by **Prof. Usman**

Thermodynamics

Fluid Mechanics

Machine Design

CAD / CAM

Manufacturing Processes

Heat Transfer

Mechanics of Materials

Engineering Dynamics

TECHNICAL SKILLS

CAD & Design

- SOLIDWORKS (3D / 2D Modeling)
- Parametric & Advanced Surface Modeling
- Technical Drawing with GD&T
- Photorealistic Rendering

Simulation Software

- ANSYS (Structural & Thermal)
- Altair EDEM (DEM Simulation)
- SolidWorks Simulation & Flow
- Motion Analysis

Engineering Analysis

- Finite Element Analysis (FEA)
- Computational Fluid Dynamics (CFD)
- Design for Assembly (DFA)
- Design for Manufacturability (DFM)
- Minitab (Data Analysis)

Manufacturing & Prototyping

- Additive Manufacturing (3D Printing)
- CNC Machining Optimization
- Injection Molding Prep
- Tolerance Stack-up Analysis

COURSES & CERTIFICATIONS

- ♦ **IELTS Academic** — Band 6.0 (Listening, Reading, Writing, Speaking)
- ♦ **Certified SOLIDWORKS Associate (CSWA)** — Dassault Systèmes
- ♦ **Pakistan Engineering Council (PEC)** — Registered Engineer
- ♦ **Lean Six Sigma Yellow Belt (LSSYB)** — udemy.com
- ♦ **Lean Six Sigma White Belt (LSSWB)** — udemy.com
- ♦ **Machine Learning in Python** — codanics.com
- ♦ **Introduction to Minitab** — simplilearn.com
- ♦ **CPR, AED & FIRST AID** — alison.com
- ♦ **ABC of Statistics** — codanics.com
- ♦ **AI ka Chilla 2026** — codanics.com
- ♦ **Python** — codanics.com

FEATURED PROJECTS

1. Fitness Product Development — Mechanical Dispensing System

Industry: Fitness & Consumer Products

Key Contributions:

- Designed mechanical assemblies involving gears, rack mechanisms, springs, sliding mechanisms, and functional interfaces.

- Developed solutions for multi-function operation within limited mechanical movement constraints.
- Performed multiple design iterations to improve mechanism reliability and manufacturability.
- Applied DFM and DFA principles for production-oriented design.
- Conducted motion analysis and mechanical validation.

Challenges Managed:

Optimizing complex mechanical interactions while maintaining compact design requirements. Achieved functional mechanism performance through iterative CAD refinement and simulation.

Tools Used:

SolidWorks (3D Modeling, Assembly Design, Motion Analysis), Altair EDEM (Simulation)

2. Medical Device Development — Controlled Flow Mechanical System

Industry: Medical Technology

Key Contributions:

- Designed mechanical components for controlled flow and operational safety.
- Developed assemblies involving valves, filters, actuators, and flow-control components.
- Applied mechanical design principles related to pressure management, thermal considerations, and fluid behavior.
- Implemented DFM and DFA methodologies for manufacturing readiness.
- Performed simulation-based design validation.

Challenges Managed:

Maintaining reliable mechanical operation under strict performance requirements. Improved design reliability through engineering analysis and simulation.

Tools Used:

SolidWorks (CAD Modeling & Motion Analysis), ANSYS (Simulation & Validation)

3. Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Industry: Academic Research (Final Year Project)

Key Contributions:

- Designed the complete 3D CAD model for the automated mechanism.
- Developed a comprehensive solution to automate the manual lifting process, significantly reducing human effort.
- Engineered a customized lifting-to-adjustment sequence for seamless water bottle installation and atomization.
- Optimized the design by utilizing a minimum number of mechanisms to achieve cost-effective and reliable results.

Challenges Managed:

Designing a reliable lifting mechanism capable of handling 18-liter loads safely and efficiently. Balancing cost-effectiveness with robust mechanical performance through design optimization.

Tools Used:

SolidWorks (3D Modeling, Simulation, Animation, Motion Analysis)

AI FEATURED
PROJECTS

Doc Agent — AI Research Assistant

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- Python
- Streamlit
- Large Language Models (LLMs)
- Document Parsing

LANGUAGES

Language	Level
English	Proficient (B2)
Urdu	Fluent (C2)
Seraiki	Native (C2)

ACADEMIC
REFERENCES

Prof. Dr. Naveed Ahmad

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